

Model Training Configuration

FineWeb-Edu Pilot Training — configuration extracted from the provided notebook

1. Model Architecture

Parameter	Value	Notes
Vocabulary size	40,000	QWEN3_CONFIG
Embedding rank	256	Factorized embedding
Context length	8,192	Model configuration
Embedding dimension	640	Transformer hidden size
Number of layers	20	Transformer blocks
Attention heads	10	Query heads
KV groups	5	Grouped-query attention
Head dimension	64	Per attention head
FFN hidden dimension	2,560	Feed-forward dimension
QK normalization	True	Enabled
RoPE base	1,000,000	RoPE frequency base
Total parameters	133,476,480	Observed from notebook
Random seed	123	PyTorch/model initialization

2. Dataset & Input Pipeline

Parameter	Value	Notes
Dataset	HuggingFaceFW/fineweb-edu	Hugging Face dataset
Subset	sample-10BT	Streaming subset
Sequence length	2,048	Training sequence length
Physical batch size	1	One sequence per DataLoader batch
Gradient accumulation	4	Four physical batches per optimizer update
Effective batch size	4 sequences	Derived from 1×4
Training token target	1,000,000,000	1B tokens
Validation token target	5,000,000	5M tokens
Validation holdout	10,000 documents	Taken from stream
Shuffle buffer	10,000	Streaming shuffle
DataLoader	drop_last=True	Train and validation
Pin memory	CUDA dependent	Enabled when CUDA is available

3. Optimization & Training

Parameter	Value	Notes
Epochs	1	Pilot run
Optimizer	AdamW	PyTorch AdamW
Learning rate	3e-4	Initial LR
Weight decay	0.01	AdamW

LR scheduler	CosineAnnealingLR	Cosine schedule
Scheduler T_max	steps_per_epoch × epochs	Computed from token target
Gradient clipping	max_norm = 1.0	Applied after unscaling
Mixed precision	GradScaler	Enabled when CUDA is available
Loss	Cross entropy	Next-token prediction
Perplexity	exp(loss)	Loss capped at 20 before exp

4. Training-Step Mechanics

Stage	Setting	Purpose
Forward	Batch size 1 × seq 2048	Generate logits
Loss	Cross entropy	Next-token prediction
Loss scaling	loss / 4	Correct gradient accumulation
Backward	GradScaler	Mixed-precision backpropagation
Optimizer update	Every 4 physical steps	Effective batch = 4
Gradient unscale	Before clipping	Correct gradient norm
Gradient clipping	1.0	Numerical stability
Scheduler step	After optimizer step	Update learning rate
Zero gradients	After update	Start next accumulation cycle

5. MLflow Configuration & Tracking

Item	Value	Purpose
Experiment	fineweb-edu-pilot-training	Groups related runs
Run name	fineweb-edu-pilot-run	Identifies this run
Parameter logging	Enabled	Stores model/training configuration
Metric logging	Every optimizer step	Avoids microbatch spam
Artifact logging	Checkpoint .pt	Associates model state with run

MLflow Parameters Logged

Category	Metrics / parameters	Purpose
Model	vocab size, emb dim, layers, heads, KV groups	Record architecture
Scale	total parameters	Record model size
Optimization	optimizer, learning rate, accumulation steps	Record training setup
Data	dataset, subset, seq length	Record data setup
Token budget	train and validation token targets	Record experiment budget

6. MLflow Runtime Metrics

Category	Metrics	What it tells you
Training	loss, perplexity	Learning progress
Optimization	learning rate	Current LR schedule
Weights	global weight norm, embedding norm	Parameter magnitude
Gradients	global grad norm, embedding grad norm	Optimization behavior
Updates	update norm, update/weight ratio	Size of parameter changes
Stability	NaN count, Inf count, GradScaler scale	Numerical health
Throughput	tokens/sec	Training speed
GPU	allocated/reserved memory	Memory usage
Layer-level	attention, MLP, norm, residual metrics	Internal model diagnostics

7. Validation Tracking

Metric	When	Purpose
Validation loss	End of each epoch	Generalization signal
Validation perplexity	End of each epoch	Interpretable LM metric

8. Checkpoint Configuration

Checkpoint field	Stored value	Why it matters
model	model.state_dict()	Learned model weights
optimizer	optimizer.state_dict()	AdamW optimizer state
scheduler	scheduler.state_dict()	Current LR schedule state
scaler	scaler.state_dict()	Mixed-precision scaler state

config	QWEN3_CONFIG	Architecture configuration
File	model_fineweb_edu_pilot.pt	PyTorch checkpoint
MLflow artifact	Checkpoint file	Associates checkpoint with experiment

9. Important Checkpoint Limitation

The notebook checkpoint stores model, optimizer, scheduler, scaler and model configuration. It does not explicitly store epoch/step counters, RNG states, or an exact streaming dataloader position. Therefore, it is a useful training-state checkpoint, but not a fully deterministic exact-resume checkpoint.

10. Inference

Parameter	Value	Notes
Mode	model.eval()	Evaluation mode
Generation	Autoregressive	Token-by-token generation
Max new tokens	200	Pilot inference
EOS token	tokenizer.eos_token_id	Stops generation when encountered

Source: Final pilot training.ipynb.txt provided in this conversation. Values above are taken from the notebook rather than inferred from external sources.